

Tools and APIs

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Conversation State and Memory

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Tools and API contracts. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Tools and API contracts. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A tool can calculate or search supplied notes. The request is 'What is 17×6 , and where do roots take up water?' Assign each subtask and describe output checks.

6. Solve the given case

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Allow only retrieval and calculator tools with validated arguments and visible results.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A tool can calculate or search supplied notes. The request is 'What is 17×6 , and where do roots take up water?' Assign each subtask and describe output checks.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Proposed call lookup_room has room_id=B and extra argument book=true. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Tools and APIs

Check a library lookup call before it reaches a tool.

lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Tools and APIs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check shape

Exactly book_id is supplied, and C7 is text.



2. Check membership

C7 occurs in the authorised registry.



3. Read only

Run the lookup. This tool contract contains no lending or deletion operation.

CHANGE ONE THING

Add lend: true to the same call.

Explain the changed result using the matching lesson.

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check a library lookup call before it reaches a tool.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run the lookup. This tool contract contains no lending or deletion operation.

1. Check shape
2. Read only
3. Check membership

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Read only Run the lookup. This tool contract contains no lending or deletion operation. Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Add lend: true to the same call. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Reject the call before invoking the tool.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

Repair a missing building block

Review Conversation State and Memory (ai-3/20). Goal: Explain the mechanism and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the mechanism and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the mechanism and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
